ARMORER'S BENCH TECHNICAL MANUAL

Smith & Wesson K/L/N-Frame Revolvers

Smith & Wesson • Distinguished Combat Magnum 686, 586, 19, 10 & Model 29 (.357 Mag, .44 Mag, .38 Spl)

Manufacturer:	Smith & Wesson	**Category:**	Revolvers
Action Type:	Double-Action / Single-Action Revolver	**Era / Service:**	American Revolver Benchmark (1899–Present)
Calibers:	.357 Magnum, .44 Magnum, .38 Special (+P)	**Covered Models:**	Model 686 / 586 (L-Frame .357), Model 19 / 66 (K-Frame Combat Magnum), Model 10 / 64 (K-Frame Service), Model 29 / 629 (N-Frame .44 Mag), Model 27 / 28 Highway Patrolman

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Cylinder Endshake Shimming (0.001"–0.002") • Barrel-Cylinder Gap Feeler Gauging (0.004"–0.007") • Rebound Slide Spring Stoning & Tuning • Crane / Yoke True Alignment • Hand Timing & Extractor Star Check

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
BARREL CYLINDER GAP	0.004" to 0.007" measured with cylinder pushed fully rearward against recoil shield
CYLINDER ENDSHAKE	0.001" to 0.002" maximum allowable fore-and-aft movement; shim with Power Custom shims if exceeding 0.003"
HEADSPACE RIMMED	0.060" GO / 0.068" NO-GO between cylinder rear face and recoil shield
TRIGGER PULL DA	9.5 to 11.0 lbs (factory standard) / 7.5 to 8.5 lbs (tuned with lightened rebound spring)
TRIGGER PULL SA	3.0 to 3.5 lbs crisp break with zero perceptible creep
SIDEPLATE REMOVAL RULE	NEVER pry sideplate with screwdriver; rap wooden handle of hammer against grip frame until sideplate walks free under vibration
CRANE ALIGNMENT	Must pass straight into cylinder window without rubbing bottom frame window ledge

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Cylinder Binds / Heavy Trigger Drag During Rotation **Trigger becomes extremely heavy or cylinder locks solid while attempting to pull trigger in double action**	• Ejector rod unscrewed and backed out (S&W; ejector rods feature left-hand reverse threads on modern guns). • Unburned powder flakes or carbon packed beneath the star-shaped extractor ratchet. • Excessive cylinder endshake allowing cylinder face to rub against barrel breech shoulder. • Bullet walking forward out of case under heavy magnum recoil and striking frame.	**Immediate Action: Look at ejector rod. If knurled head has backed out, tighten counter-clockwise (reverse thread) by hand. Gunsmith Action: Swing out cylinder, lift extractor star, brush under-star recess with bronze brush. Check barrel-cylinder gap with feeler gauge. If gap is under 0.003", face off barrel breech shoulder with 90° facing cutter.**

Stoppage 2: Cylinder Fails to Carry-Up (Late Hand Timing) **Hammer drops or reaches full cock before the cylinder bolt snaps into the cylinder locking notch**	• Worn or peened hand tip (the vertical pawl that pushes the extractor ratchet teeth). • Worn ratchet tooth on one or more individual chambers. • Cylinder bolt window binding on dried grease or metal burr.	**Gunsmith Action: TIMING TEST: Hold cylinder lightly with thumb and forefinger to create gentle drag. Pull trigger slowly in double action. Cylinder bolt MUST snap into locking notch with an audible click BEFORE hammer falls. If late, install oversized hand and hand-stone width to time each chamber.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
S&W; Revolver Tuning, Endshake & Action Job Masterclass	Mark Novak / Anvil Gunsmithing	Masterclass on removing S&W; sideplates without damaging mating seams, checking barrel-cylinder gap, installing endshake bearings, and stoning the rebound slide.
Jerry Miculek: How to Tune a Smith & Wesson Revolver Trigger	Jerry Miculek / Miculek Pro Bench	Legendary speed shooter Jerry Miculek demonstrates mainspring strain screw adjustment, rebound spring replacement, and mirror stoning trigger contact surfaces.